

A Square-Overlap Obstruction for Unconditional Gabor Frames in $L^p(\mathbb{R})$, $1 < p < 2$

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Status

This is a likely valid partial result for the open problem in Lev–Tselishchev [1]. It does not solve the arbitrary countable-index case. It isolates a necessary condition for any counterexample to the expected nonexistence theorem.

Source Problem

The source paper studies Gabor systems

$$\mathcal{G}(g; \Lambda) = \{e_s \tau_t g : (t, s) \in \Lambda\}, \quad (e_s \tau_t g)(x) = g(x - t) e^{2\pi i s x},$$

in $L^p(\mathbb{R})$, $p \neq 2$. For $1 < p < 2$, Theorem 2.4 of [1] proves nonexistence of unconditional Gabor bases and unconditional Schauder frames when $\Lambda \subset T \times \mathbb{R}$ for a uniformly discrete, more generally bounded-density, time set T . The paper then leaves open the corresponding nonexistence question with no a priori condition on Λ or g .

set $\Lambda \subset \mathbb{R}$. Then no system (2.1) can form an unconditional Schauder frame in $L^p(\mathbb{R})$. In particular, there is no unconditional basis of the form $\mathcal{G}(g; \Lambda)$ in $L^p(\mathbb{R})$.

Moreover, we prove a result about unconditional Gabor basic sequences: if the Gabor system $\mathcal{G}(g; \Lambda)$ forms an unconditional basic sequence in $L^p(\mathbb{R})$, $1 \leq p < 2$, and if $\Lambda \subset T \times \mathbb{R}$ for some uniformly discrete set $T \subset \mathbb{R}$, then the system is equivalent to a subsequence of the standard unit vector basis of $(\ell^2 \oplus \ell^2 \oplus \cdots)_p$. As a consequence, the system $\mathcal{G}(g; \Lambda)$ cannot span $L^p(\mathbb{R})$. Note that in this result (contrary to the $p > 2$ case) we do not assume that the system $\mathcal{G}(g; \Lambda)$ spans a complemented subspace of $L^p(\mathbb{R})$.

The results are true more generally in the case where $T \subset \mathbb{R}$ is a finite union of uniformly discrete sets. On the other hand, we will present an example showing that our result about unconditional Gabor basic sequences in $L^p(\mathbb{R})$, $1 \leq p < 2$, becomes false if the uniform discreteness assumption is dropped (see Section 4).

We suspect that no unconditional Gabor bases or frames exist in $L^p(\mathbb{R})$, $1 < p < 2$, without imposing any a priori assumptions on the countable set $\Lambda \subset \mathbb{R} \times \mathbb{R}$ or the function $g \in L^p(\mathbb{R})$; however we leave this problem open.

2.5. In the remainder of the paper, we prove the theorems stated above and provide some additional remarks about the results. In Section 3, we review some preliminary basic facts from Banach space theory and Fourier analysis, which are used throughout the paper. In Sections 4 and 5 we prove our results on unconditional Gabor basic sequences and frames in $L^p(\mathbb{R})$ spaces, $1 < p < 2$. In Sections 6, 7 and 8, the results concerning the case $p > 2$ are presented.

Remark. The results extend, with essentially the same proofs, to higher dimensions, i.e., they hold for Gabor systems in $L^p(\mathbb{R}^d)$ for any dimension d . We present the results in dimension one merely in order to simplify the exposition.

Result

For a countable $\Lambda \subset \mathbb{R}^2$, write

$$T = \{t \in \mathbb{R} : \text{there is } s \in \mathbb{R} \text{ with } (t, s) \in \Lambda\}, \quad S_t = \{s : (t, s) \in \Lambda\}.$$

Theorem 1 (Square-overlap obstruction). *Let $1 < p < 2$, let $0 \neq g \in L^p(\mathbb{R})$, and let $\Lambda \subset \mathbb{R}^2$ be countable. Suppose that*

$$H_2(x) := \left(\sum_{t \in T} |g(x - t)|^2 \right)^{1/2}$$

is finite on a set of positive Lebesgue measure. Then no choice of coefficient functionals $g_{t,s}^ \in (L^p(\mathbb{R}))^*$ can make*

$$\{(e_s \tau_t g, g_{t,s}^*)\}_{(t,s) \in \Lambda}$$

an unconditional Schauder frame for $L^p(\mathbb{R})$. Consequently no Gabor system $\mathcal{G}(g; \Lambda)$ satisfying this square-overlap hypothesis is an unconditional basis for $L^p(\mathbb{R})$.

Corollary 1. *If an unconditional Gabor Schauder frame exists in $L^p(\mathbb{R})$, $1 < p < 2$, then for its time projection T one must have*

$$\sum_{t \in T} |g(x - t)|^2 = \infty$$

almost everywhere on every measurable set of positive measure.

Proof Intuition

The proof keeps the coefficient-estimate half of the source argument and changes only the final obstruction. Unconditionality and cotype 2 give a uniform bound from $L^p(\mathbb{R})$ into local L^2 -valued coefficient functions $(u_t f)_{t \in T}$. If the translated copies of the window are square-summable on a positive-measure set, then the frame expansion gives the pointwise estimate

$$|f(x)| \leq H_2(x) \left(\sum_t |u_t f(x)|^2 \right)^{1/2}.$$

On a subset where H_2 is bounded, this would make the inclusion $L^p(E) \hookrightarrow L^2(E)$ bounded. Since $1 < p < 2$, that is impossible on any positive-measure Lebesgue set.

Proof

We use two standard ingredients that are also used in Sections 3 and 5 of [1]: unconditional Schauder frames have a uniform unconditional constant, and $L^p(\mathbb{R})$, $1 < p < 2$, has cotype 2. We also use the nonuniform Bessel estimate for exponentials stated as Lemma 3.7 in [1].

Lemma 1 (Block coefficient estimate). *Assume that $\{(e_s \tau_t g, g_{t,s}^*)\}_{(t,s) \in \Lambda}$ is a K -unconditional Schauder frame for $L^p(\mathbb{R})$, $1 < p < 2$, and normalize $\|g\|_p = 1$. There are $\delta > 0$ and $C < \infty$ such that, for*

$$S_{t,l} = S_t \cap [l\delta, (l+1)\delta), \quad l \in \mathbb{Z},$$

every $f \in L^p(\mathbb{R})$ satisfies

$$\left(\sum_{t \in T} \sum_{l \in \mathbb{Z}} \left(\sum_{s \in S_{t,l}} |g_{t,s}^*(f)| \right)^2 \right)^{1/2} \leq C \|f\|_p.$$

Proof. Choose $\delta > 0$ so small that

$$\|(e_a - 1)g\|_p < 1/2, \quad |a| < \delta,$$

which is possible by dominated convergence. For finite collections of time-frequency blocks, set $c_{t,s} = g_{t,s}^*(f)$ and

$$y_{t,l} = \sum_{s \in S_{t,l}} |c_{t,s}| e^{-2\pi i s t} e_s \tau_t g.$$

The unconditional frame estimate, applied with scalar multipliers of modulus at most one, gives

$$\mathbb{E} \left\| \sum_{t,l} \varepsilon_{t,l} y_{t,l} \right\|_p \leq K \|f\|_p.$$

Cotype 2 therefore yields

$$\left(\sum_{t,l} \|y_{t,l}\|_p^2 \right)^{1/2} \leq C_p K \|f\|_p.$$

If $S_{t,l} \neq \emptyset$, choose $s_{t,l} \in S_{t,l}$. For $s \in S_{t,l}$,

$$\|e^{-2\pi i s t} e_s \tau_t g - e^{-2\pi i s_{t,l} t} e_{s_{t,l}} \tau_t g\|_p = \|(e_{s-s_{t,l}} - 1)g\|_p < 1/2.$$

By the triangle inequality,

$$\|y_{t,l}\|_p \geq \frac{1}{2} \sum_{s \in S_{t,l}} |c_{t,s}|.$$

Combining the last two displays and passing from finite subcollections to the countable family proves the estimate. $\square$

Lemma 2 (Local L^2 analysis estimate). *Under the hypotheses of Lemma 1, for each $t \in T$ define*

$$(u_t f)(x) = \sum_{s \in S_t} g_{t,s}^*(f) e_s(x).$$

Then the series converges unconditionally in $L^2(I)$ for every unit interval I , and there is $C_I < \infty$, independent of f , such that

$$\sum_{t \in T} \|u_t f\|_{L^2(I)}^2 \leq C_I \|f\|_p^2.$$

Proof. Apply the nonuniform exponential Bessel estimate of [1, Lemma 3.7] to each frequency set S_t , using the same δ as in Lemma 1. For every unit interval I ,

$$\|u_t f\|_{L^2(I)} \leq C(\delta) \left(\sum_{l \in \mathbb{Z}} \left(\sum_{s \in S_{t,l}} |g_{t,s}^*(f)| \right)^2 \right)^{1/2}.$$

Squaring, summing in t , and invoking Lemma 1 gives the claim. $\square$

We now prove Theorem 1. Suppose that an unconditional Schauder frame as above exists. Since $H_2 < \infty$ on a set of positive measure, there are a bounded interval I , a measurable set $E \subset I$ with $m(E) > 0$, and a finite constant M such that $H_2(x) \leq M$ for almost every $x \in E$.

Let $f \in L^p(\mathbb{R})$ be supported on E . The unconditional frame expansion may be grouped first by time:

$$f = \sum_{t \in T} \sum_{s \in S_t} g_{t,s}^*(f) e_s \tau_t g.$$

For each fixed t , the inner subseries agrees locally a.e. with

$$g(x - t)(u_t f)(x).$$

By Lemma 2, $(u_t f(x))_{t \in T} \in \ell^2(T)$ for a.e. $x \in I$. On E , the vector $(g(x - t))_{t \in T}$ is also in $\ell^2(T)$. Hence the grouped time series is absolutely convergent for a.e. $x \in E$. Passing to pointwise convergent subsequences of the unconditional partial sums identifies its value with $f(x)$, and Cauchy's inequality gives

$$|f(x)| \leq \sum_{t \in T} |g(x - t)| |u_t f(x)| \leq H_2(x) \left(\sum_{t \in T} |u_t f(x)|^2 \right)^{1/2}.$$

Consequently Lemma 2 gives

$$\|f\|_{L^2(E)}^2 \leq M^2 \int_I \sum_{t \in T} |u_t f(x)|^2 dx \leq M^2 C_I \|f\|_p^2.$$

Thus every L^p -function supported on E would belong to $L^2(E)$, with a uniform bound $\|f\|_2 \lesssim \|f\|_p$. This is impossible for $1 < p < 2$ on a positive-measure Lebesgue set: choose measurable subsets $E_n \subset E$ with $0 < m(E_n) \rightarrow 0$, and put

$$f_n = m(E_n)^{-1/p} \mathbf{1}_{E_n}.$$

Then $\|f_n\|_p = 1$ but $\|f_n\|_2 = m(E_n)^{1/2-1/p} \rightarrow \infty$. The contradiction proves the theorem. The assertion about unconditional bases follows because an unconditional basis induces an unconditional Schauder frame with its coordinate functionals.

Relation to the Source Theorem

If T has bounded density, then for every bounded interval I ,

$$\int_I \sum_{t \in T} |g(x-t)|^p dx < \infty.$$

Hence $(g(x-t))_{t \in T} \in \ell^p(T) \subset \ell^2(T)$ for a.e. $x \in I$. Theorem 1 therefore recovers the $1 < p < 2$ bounded-density frame nonexistence theorem of [1]. The present formulation additionally covers arbitrary countable time projections whenever their translated-window square overlap is finite on a set of positive measure.

Limitations and Novelty Check

This packet does not rule out the remaining dense-overlap case

$$\sum_{t \in T} |g(x-t)|^2 = \infty \quad \text{a.e.}$$

That is the exact surviving obstruction after this argument. The source paper contains the coefficient estimates used above but states the $1 < p < 2$ nonexistence theorem under bounded-density hypotheses. A bounded search of the run indexes and web literature for the exact Gabor L^p open problem, the source arXiv id, and the related exponential-frame paper [2] found no later full solution of the arbitrary countable Gabor case. Novelty confidence is therefore moderate: the result is best viewed as an extracted and sharpened partial obstruction from the source method, not as an independent full solution.

References

- [1] N. Lev and A. Tselishchev, *Gabor unconditional bases and frames in $L^p(\mathbb{R})$* , arXiv:2605.17970, 2026.
- [2] N. Lev and A. Tselishchev, *Unconditional Schauder frames of exponentials and of uniformly bounded functions in L^p spaces*, International Mathematics Research Notices, 2025, no. 19, rnaf299.